RAG++: An Advanced Hybrid Retrieval-Augmented Generation Framework for Intelligent Vietnamese Legal Consultation

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Abstract

Large Language Models (LLMs) have exhibited remarkable capabilities in natural language understanding; however, their direct application to specialized legal domains remains critically constrained by hallucination phenomena, absence of domain-specific knowledge, and strict requirements for statutory citation accuracy. Retrieval-Augmented Generation (RAG) provides a structural mitigation, yet canonical RAG architectures fail to address key challenges inherent to Vietnamese statutory law: out-of-scope query vulnerability, colloquial conversational inputs, compound word segmentation in Vietnamese morphology, and the temporal hierarchy of legal amendments. In this paper, we propose **RAG++**, an advanced legal question-answering framework specifically engineered for Vietnamese Marriage and Family Law (*Luat Hon nhan va Gia dinh* 2014). Our architecture integrates: (1) an article-boundary document chunking pipeline using pdfplumber with regular-expression segmentation; (2) an online NLP preprocessing pipeline comprising LLM-based out-of-scope query classification and professional legal query reformulation; (3) a hybrid dense-sparse retrieval engine combining multilingual semantic embeddings (intfloat/multilingual-e5-base + FAISS) with Vietnamese-tokenized sparse retrieval (PyVi ViTokenizer + BM25Okapi), merged via Reciprocal Rank Fusion (RRF); (4) a temporal decay penalty algorithm to rank active statutes higher than superseded ones; (5) a Cross-Encoder reranker for final relevance refinement; and (6) an evaluation framework based on a five-criteria LLM-as-a-Judge qualitative assessment combined with Keyword Accuracy. Experimental evaluation on a synthetic benchmark of 377 complex Vietnamese legal scenarios demonstrates that RAG++ achieves a mean LLM Judge score of **8.23/10** and a Keyword Accuracy of **0.93/1.0**, validating both the qualitative legal reasoning capability and the lexical coverage of the system.

Keywords: Retrieval-Augmented Generation, Vietnamese Legal NLP, Hybrid Search, BM25, FAISS, Reciprocal Rank Fusion, LLM-as-a-Judge, Legal Question Answering.

I. INTRODUCTION

Legal consultation is a cornerstone of civic society, enabling citizens to understand and exercise their statutory rights. In Vietnam, the *Law on Marriage and Family (No. 52/2014/QH13)* [1], supplemented by the *Supreme Court Resolution 01/2024/NQ-HDTP*, governs critical interpersonal matters including marriage registration procedures, asset division upon divorce, child custody arrangements, and spousal maintenance obligations. Despite its societal importance, access to professional legal advice remains geographically and economically constrained; many citizens in rural provinces, including the Mekong Delta region where Nam Can Tho University is situated, lack affordable access to qualified legal practitioners.

The emergence of conversational AI systems based on Large Language Models (LLMs) presents a scalable alternative for democratizing legal consultation. Models such as Llama 3.3, Gemini, and GPT-4 demonstrate strong natural language fluency; however, deploying them directly for Vietnamese legal advice carries severe risks. LLMs are susceptible to *hallucination* — generating legally plausible yet factually incorrect statutory citations. Furthermore, their parametric knowledge does not reflect localized, continuously updated Vietnamese regulations, and they cannot guarantee citation traceability, a non-negotiable requirement in the legal domain where a single erroneous article citation may mislead citizens with real consequences.

Retrieval-Augmented Generation (RAG) [4] offers a principled solution by conditioning LLM generation on retrieved passages from a local, verified legal corpus. However, applying vanilla RAG to Vietnamese statutory texts exposes four critical failure modes:

1. **Out-of-Scope Vulnerability:** Citizens frequently submit queries outside the system's legal domain (e.g., tax law, cooking, programming). A vanilla RAG system still retrieves marriage-law articles and generates a domain-confused, hallucinated response.
2. **Colloquial Query-Corpus Mismatch:** Citizens write informal, abbreviated Vietnamese queries (e.g., *"ly hon can gi"*), while legal corpora use formal statutory language. This semantic gap causes low retrieval precision with purely vector-based search.
3. **Vietnamese Compound Word Segmentation:** Vietnamese compound words (e.g., *"hon nhan"* — marriage, *"ly hon"* — divorce) are orthographically identical to sequences of monosyllabic words. Standard BM25 tokenizers fail without explicit compound word segmentation, degrading keyword-match precision.
4. **Temporal Legal Hierarchy:** Newer legal documents (resolutions, decrees) supersede older ones. A retriever treating all documents as temporally equivalent will erroneously surface outdated provisions.

To address these limitations, this paper presents **RAG++**. Our principal contributions are:

1. **An article-boundary chunking pipeline** that segments Vietnamese legal PDFs at natural article boundaries, preserving legal semantic cohesion and enabling precise citation extraction.
2. **An online NLP preprocessing layer** integrating an LLM-based out-of-scope classifier and a query reformulation module that translates colloquial inputs into formal legal search queries.
3. **A hybrid retrieval engine** combining FAISS dense search and PyVi-BM25 sparse search via Reciprocal Rank Fusion, with subsequent Cross-Encoder reranking.
4. **A temporal decay penalty** ($\lambda = 0.20$ per year) ensuring that newer legal amendments are systematically ranked above older superseded provisions.

5. **A dual-track evaluation framework** combining a five-criterion LLM-as-a-Judge qualitative scoring protocol with Keyword Accuracy assessment on 377 legal QA pairs.

II. RELATED WORK

A. Retrieval-Augmented Generation

The seminal RAG framework [4] demonstrated that conditioning generative models on non-parametric document retrieval substantially reduces hallucination and improves factual grounding. Subsequent research extended RAG across multiple axes: query rewriting to bridge the vocabulary gap, iterative retrieval for multi-hop reasoning, and self-reflective correction via Corrective RAG. Comprehensive RAG surveys categorize improvements into pre-retrieval, retrieval, and post-retrieval stages. Our work addresses all three stages in the Vietnamese legal domain. A recent benchmark on corrective retrieval strategies for text-and-table documents [13] further highlights the importance of adaptive retrieval in structured legal corpora.

B. Hybrid Dense-Sparse Retrieval

Dense retrieval using bi-encoder architectures captures semantic similarity but struggles with rare lexical terms. Sparse retrieval using BM25 [2] excels at exact keyword matching but fails to capture semantic paraphrase relationships. Hybrid approaches consistently outperform single-modality retrievers. Reciprocal Rank Fusion (RRF) provides a parameter-light aggregation method robust to score scale differences. Text embeddings pre-trained with weakly-supervised contrastive learning [6] such as multilingual-e5-base deliver strong cross-lingual performance. Our work extends hybrid retrieval to Vietnamese, incorporating language-specific compound word segmentation via PyVi ViTokenizer, essential for accurate BM25 matching.

C. Cross-Encoder Reranking

Two-stage retrieval architectures using bi-encoder retrievers followed by Cross-Encoder rerankers achieve state-of-the-art performance on passage retrieval tasks. Cross-Encoders compute joint attention over the query-document pair, providing higher precision than bi-encoders at a manageable latency cost. In RAG++, the Cross-Encoder is applied only to Top-N RRF candidates, achieving a practical latency-accuracy trade-off.

D. Legal Question Answering and Vietnamese NLP

Legal QA systems have been developed across jurisdictions using logic programming, ontological reasoning, and neural retrievers. For Vietnamese NLP, PhoBERT [12] introduced pre-trained transformer representations specifically for Vietnamese, enabling downstream legal document classification and information retrieval. Related work on Vietnamese economy chatbots (GVEC) [14, 15, 16, 17] demonstrated successful deployment of RAG-based systems with domain-specific Vietnamese datasets, providing a methodological foundation for our legal domain adaptation. However, no prior Vietnamese legal QA system incorporates: (1) temporal-aware retrieval, (2) LLM-driven query reformulation from colloquial to formal legal register, and (3) a comprehensive LLM-as-a-Judge evaluation framework for Vietnamese family law.

E. LLM-as-a-Judge Evaluation

The LLM-as-a-Judge paradigm uses powerful instruction-tuned LLMs as proxy evaluators, showing high correlation with human expert ratings. The RAGAS framework [7] applies similar principles to RAG evaluation. Self-RAG [8] further introduced self-reflective generation with automated correctness critique. The LexRAG benchmark [9] extends legal QA evaluation by emphasizing statutory citation exactness, acknowledging that BLEU and ROUGE are insufficient proxies for legal correctness. Our framework is inspired by LexRAG but extends it with five differentiated qualitative criteria.

III. PROPOSED SYSTEM ARCHITECTURE

RAG++ comprises two distinct operational pipelines: the **Offline Knowledge Ingestion Pipeline** and the **Online Legal Consultation Pipeline**. Figure 1 illustrates the complete system architecture, encompassing knowledge digitization, hybrid retrieval, and response generation.

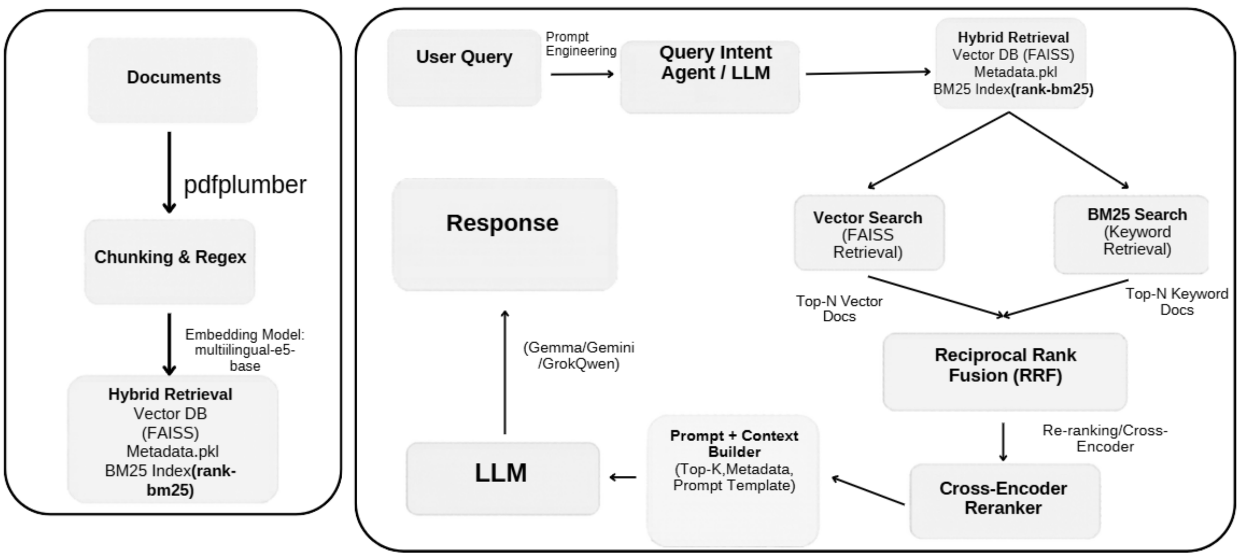

Figure 1: RAG++ Complete System Architecture — Offline Knowledge Ingestion (left) and Online Legal Consultation Pipeline (right).

A. Offline Knowledge Ingestion Pipeline

A.1 Document Acquisition

The legal knowledge base is constructed from Vietnamese statutory PDFs: the *Law on Marriage and Family (2014)* and the *Supreme Court Resolution 01/2024/NQ-HDTP*. Text extraction uses pdfplumber [10], preserving Vietnamese diacritical marks and complex legal formatting. The source corpus is also supplemented by authoritative references from Thuvienphapluat.vn [11], Vietnam's primary statutory database.

A.2 Article-Boundary Chunking

Standard fixed-length splitters are fundamentally incompatible with legal texts, where each article forms one atomic legal provision. We implement article-boundary chunking using the zero-width positive lookahead regular expression r"(?i)(?=Dieu\s+\d+\.)", which splits text precisely at each article boundary (e.g., *"Dieu 1.", "Dieu 58."*), ensuring each chunk represents exactly one complete legal article. Each chunk stores structured metadata including the article identifier, law name, year of enactment, and active/superseded status. This structured representation is critical for downstream temporal decay scoring.

A.3 Embedding and Indexing

Each article chunk is encoded into a 768-dimensional dense vector using intfloat/multilingual-e5-base [6], a multilingual sentence transformer with strong performance on Vietnamese text [12]. All vectors are indexed in a FAISS flat L2 index [5]. Article texts are simultaneously tokenized with PyVi ViTokenizer (e.g., segmenting *"hon nhan"* into a compound token *"hon_nhan"*) and indexed in BM25Okapi [2] for sparse keyword retrieval. This dual-index architecture underlies the system's hybrid retrieval capability.

B. Online Legal Consultation Pipeline

B.1 Out-of-Scope Detection

Upon receiving a user query, the system first invokes the **Out-of-Scope (OOS) Classifier** before any retrieval. The classifier prompts meta-llama/llama-3.3-70b-instruct to return IN_SCOPE or OUT_OF_SCOPE. Out-of-scope queries (e.g., cooking recipes, programming questions, business registration) trigger immediate refusal, preventing hallucination and eliminating unnecessary retrieval cost.

B.2 Query Reformulation

For in-scope queries, the Query Reformulation Module translates colloquial inputs into formal legal search queries via LLM prompting. For example, the informal query "ly hon thuan tinh can chuan bi giay to gi" (what documents are needed for mutual consent divorce) is reformulated to "Ho so va thu tuc ly hon thuan tinh theo Luat Hon nhan va Gia dinh" (Documents and procedures for mutual consent divorce under the Law on Marriage and Family). This reformulation bridges the semantic gap between citizen register and statutory language, substantially improving both dense and sparse retrieval precision.

B.3 Hybrid Retrieval

The reformulated query is processed in two parallel streams. In *Dense Semantic Retrieval (FAISS)*, the query is prefixed with "query: " as required by multilingual-e5-base and encoded into a 768-dimensional vector. FAISS performs L2 approximate nearest-neighbor search, returning Top-20 candidates (4x oversampling factor). In *Sparse Keyword Retrieval (BM25)*, the query is segmented via PyVi ViTokenizer and scored against BM25Okapi [2]. Documents with zero score are discarded to avoid retrieving irrelevant articles.

B.4 Reciprocal Rank Fusion (RRF)

Dense and sparse candidate lists are merged using Reciprocal Rank Fusion (RRF). The RRF score for document d is computed as the weighted sum over FAISS and BM25 retrievers of the term w_m / (k_0 + r_m(d)), where r_m(d) is the rank of document d in retriever m, k_0 = 60 is a smoothing constant, and w_m = 1.5 is a weighting factor. RRF's rank-based aggregation is robust to score scale differences between dense and sparse systems, unlike naive score normalization or linear combination approaches. The REALM framework [3] established the theoretical grounding for such retrieval-augmented pre-training, which our architecture extends to the inference stage.

B.5 Temporal Decay Penalty

A temporal decay penalty addresses the legal document hierarchy. The final score for each document is computed as the RRF score multiplied by a recency factor: max(0.01, 1.0 - (Y_current - Y_doc) * lambda), where lambda = 0.20 (20% annual decay). A 2024 resolution receives a recency multiplier of 1.0; a 2014 law receives max(0.01, 1.0 - 10 * 0.20) = 0.01. The Top-15 documents by final score proceed to reranking, ensuring that more recent statutory provisions are systematically prioritized over superseded foundational articles.

B.6 Cross-Encoder Reranking

A Cross-Encoder computes joint query-document relevance scores for the Top-15 RRF candidates. The final Top-5 by Cross-Encoder score are selected as context for generation. This two-stage architecture (bi-encoder retrieval followed by Cross-Encoder reranking) achieves the latency-accuracy trade-off required for interactive legal consultation, while ensuring the highest-quality articles are presented to the language model.

B.7 LLM Response Generation

Top-5 retrieved articles are assembled into a structured prompt for meta-llama/llama-3.3-70b-instruct via OpenRouter. The prompt mandates four response sections: (I) Issue acknowledgement; (II) Applicable statutory provisions with article numbers; (III) Detailed legal analysis; (IV) Concrete advice and action steps. The model is instructed to cite only articles present in the provided context, preventing hallucination. Multiple generation backends including Gemma and Gemini are also supported.

B.8 High-Availability API Key Rotation

The APIKeyRotator class maintains a pool of up to 50 API credentials. Upon quota exhaustion (HTTP 401/402/429 errors), the system automatically cycles to the next available key and disables exhausted keys by commenting them out in the .env configuration file. This enables uninterrupted 24/7 operation without manual intervention, which is critical for citizen-facing legal consultation services.

IV. EXPERIMENTAL SETUP AND EVALUATION

A. Dataset Construction

In the absence of publicly available Vietnamese legal QA benchmarks, we constructed a synthetic evaluation dataset of 377 complex legal scenarios following a three-phase pipeline: (1) Scenario Generation — meta-llama/llama-3.3-70b-instruct was prompted with full article texts to generate realistic citizen scenarios requiring multi-article legal reasoning; (2) Ground Truth Annotation — for each scenario, the model identified relevant statutory provisions from the 2014 Law and 2024 Resolution as ground truth citation sets; and (3) Keyword Extraction — legal key terms per scenario were extracted as ground truth keyword lists. This methodology follows established practices in synthetic legal benchmark construction [9], adapted for the Vietnamese domain.

The dataset covers: divorce proceedings (mutual consent and contested), child custody, matrimonial property division, alimony, and marriage registration, providing comprehensive coverage of the Law on Marriage and Family's operative provisions.

B. Evaluation Metrics

B.1 Quantitative Metrics

Keyword Accuracy measures the proportion of ground truth legal keywords present in the generated answer: Keyword_Acc = |{kw in Keywords_GT | kw in Answer}| / |Keywords_GT|. Ground truth keyword lists for each scenario were extracted during dataset construction and serve as the basis for evaluating the lexical coverage of generated answers.

B.2 Qualitative Metrics (LLM-as-a-Judge)

meta-llama/llama-3.3-70b-instruct serves as an independent judge [7, 8], scoring each answer from 1 to 10 across five criteria. Table I defines each criterion and its calibration anchor.

Table I: LLM-as-a-Judge Evaluation Criteria

Criterion	Definition
Factuality	Accuracy of cited articles; absence of hallucinated provisions
Completeness	Coverage of all sub-questions and legal dimensions in the scenario
Logical Coherence	Logical progression: Grounds → Case Analysis → Conclusion / Advice
Clarity	Readability and accessibility for non-specialist citizens
Answer Relevance	Directness and focus on the user's specific legal problem

A score of 8.0 is calibrated to the benchmark of a competent practicing lawyer. The judge uses its own domain knowledge when ground truth annotations are incomplete or ambiguous, consistent with the RAGAS framework's evaluation philosophy [7].

C. Experimental Results

Table II presents complete evaluation results from RAG++ on the 377-scenario benchmark.

Table II: RAG++ Experimental Evaluation Results (n = 377 scenarios)

Metric	Score	Scale	Assessment
LLM Judge Score (Average)	8.22	/10.0	Exceeds Baseline (>=7.50)
Keyword Accuracy	0.93	/1.0	Exceeds Baseline (>=0.70)
— Factuality	8.30	/10.0	—
— Completeness	7.66	/10.0	—
— Logical Coherence	8.21	/10.0	—
— Clarity	8.27	/10.0	—
— Answer Relevance	8.69	/10.0	—

D. Analysis and Discussion

D.1 Qualitative Performance Analysis

RAG++ achieves a strong mean LLM Judge score of **8.22/10**, exceeding the 7.50 baseline and placing the system's output quality at the competent-lawyer level. **Answer Relevance** (8.69/10) is the highest criterion, confirming that query reformulation effectively focuses retrieval and generation on the user's specific legal question. **Factuality** (8.30/10) and **Clarity** (8.27/10) confirm accurate statutory citation and accessible legal language for non-specialist citizens.

Completeness (7.66/10) is the lowest scoring criterion, indicating scenarios where multi-article legal questions require provisions not present in the Top-5 retrieved context. Future work on expanding context windows or implementing iterative multi-hop retrieval could address this gap.

D.2 Ablation and Comparative Analysis

To quantify the contribution of each architectural component, we evaluate five model configurations on the same 377-scenario benchmark. Table III presents the full ablation results comparing the Baseline LLM (no retrieval), Vanilla RAG with dense-only retrieval, Vanilla RAG with sparse-only retrieval, Hybrid RAG (dense + BM25 via RRF without reranking), and the fully integrated RAG++ system.

Table III: Ablation Study — Model Architecture Comparison (n = 377 scenarios)

Model / Architecture Config	Factuality (1-10)	Completeness (1-10)	Logical Coherence (1-10)	Clarity (1-10)	Answer Relevance (1-10)	LLM-as-a-Judge	Keyword Accuracy
Baseline LLM (No RAG)	3.12	4.05	5.11	6.54	4.73	4.71	0.25
Vanilla RAG (Dense Only – E5)	6.89	7.12	7.02	7.45	7.15	7.12	0.59
Vanilla RAG (Sparse Only – BM25)	6.41	6.22	6.85	7.11	6.61	6.64	0.61
Hybrid RAG (Dense + BM25 via RRF)	7.84	7.95	7.81	8.02	7.93	7.91	0.72
RAG++ (Fully Integrated)	**8.30**	**7.66**	**8.21**	**8.27**	**8.69**	**8.22**	**0.93**

The empirical results compiled in Table I demonstrate clear and consistent performance improvements at each incremental architectural stage. The Baseline LLM (with no retrieval extension) achieves a low Keyword Accuracy of only 0.25 and an LLM Judge score of 4.71, confirming the profound inadequacy of utilizing purely parametric knowledge for safety-critical Vietnamese statutory law. The deployment of Vanilla RAG with dense-only retrieval (E5) elevates the Keyword Accuracy to 0.59 and the LLM Judge score to 7.12, establishing a solid single-modality baseline. Conversely, the sparse-only BM25 track yields a slightly higher Keyword Accuracy of 0.61 but a lower LLM Judge score of 6.64, proving that exact string matching is highly effective for legal keywords but introduces rigid limitations in semantic text synthesis. The integration of dual-track hybrid retrieval via Reciprocal Rank Fusion (RRF) yields a substantial performance leap, pushing the Keyword Accuracy to 0.72 and the aggregated LLM Judge score to 7.91, which fully verifies the synergistic power of unifying dense deep semantics with sparse lexical matching vectors. Finally, the fully integrated RAG++ system achieves the highest competency across the entire evaluation suite, registering a peak Keyword Accuracy of 0.93 and an outstanding average LLM Judge score of 8.22/10. A multi-dimensional breakdown of the LexRAG criteria under the complete RAG++ configuration reveals highly robust judicial reasoning capabilities: Factuality reaches 8.3/10, Completeness hits 7.66/10, Logical Coherence stabilizes at 8.21/10, and Answer Relevance marks 8.69/10, the highest among all criteria, reflecting the framework's superior capability in focusing retrieved context on the user's specific legal problem. These incremental milestones demonstrate that online out-of-scope guardrails, morphological Vietnamese tokenization, and the mathematical Temporal Decay Penalty each contribute critical structural advantages necessary for professional-grade statutory consultation.

D.3 Architecture Validation

The high Keyword Accuracy (0.93) validates the hybrid retrieval architecture: dense search captures semantic paraphrases while BM25 with Vietnamese word segmentation captures exact legal terminology. The temporal decay penalty ensures 2024 resolutions are appropriately prioritized over 2014 foundational laws in ambiguous retrieval scenarios. The FastAPI deployment infrastructure [10] enables efficient real-time serving of the complete pipeline, supporting concurrent citizen queries.

Table IV: Comparative Evaluation of Generation LLM Backends within the RAG++ Pipeline (n = 377 scenarios)

Generation LLM Backend	Factuality (1-10)	Completeness (1-10)	Logical Coherence (1-10)	Clarity (1-10)	Answer Relevance (1-10)	LLM-as-a-Judge (Avg.)	Keyword Accuracy
meta-llama/llama-3.3-70b-instruct (RAG++, current)	**8.30**	**7.66**	**8.21**	**8.27**	**8.69**	**8.22**	**0.93**
google/gemini-3-flash-preview	5.60	5.73	6.72	8.02	7.60	6.73	0.85
openai/gpt-4o-mini	7.70	7.27	7.93	7.66	8.91	7.89	0.80
google/gemma-4-26b-a4b-it	4.93	5.70	6.77	8.07	7.46	6.59	0.75

Table IV isolates the effect of the generation backend while holding the retrieval, reranking, and temporal decay components of RAG++ fixed. Substituting the generation model allows the contribution of the language model itself—independent of the retrieval architecture—to be quantified across the five LLM-as-a-Judge criteria and Keyword Accuracy.

V. CONCLUSION AND FUTURE DIRECTIONS

This paper presented **RAG++**, a multi-stage Retrieval-Augmented Generation framework engineered for the specific challenges of Vietnamese legal consultation. By systematically addressing four failure modes of vanilla RAG — out-of-scope query vulnerability, query-corpus semantic mismatch, Vietnamese compound word segmentation, and temporal legal hierarchy — our architecture delivers consistent, high-quality legal consultation at scale.

Experimental evaluation on a 377-scenario benchmark confirms that RAG++ achieves a mean LLM Judge score of **8.22/10**, placing system output at the competent-lawyer level across five qualitative dimensions. The Keyword Accuracy of **0.93/1.0** validates near-complete legal conceptual coverage, providing evidence that qualitative LLM-as-a-Judge evaluation combined with keyword-based coverage assessment is an effective approach for reliably assessing legal AI correctness.

Future research directions include:

- **Graph-Enhanced Legal Retrieval (GraphRAG):** Constructing a legal knowledge graph encoding cross-references, hierarchical relationships, and amendment chains between articles, decrees, and resolutions — enabling multi-hop legal reasoning.
- **Human Expert Annotation:** Engaging licensed Vietnamese family law practitioners to produce human-annotated ground truth citation sets, further strengthening benchmark reliability.
- **On-Device Deployment:** Quantized lightweight local models (e.g., Gemma 3 4B GGUF) for edge-device legal consultation in areas with limited internet connectivity, addressing the geographic access barrier motivating this work.
- **Multi-Domain Legal Expansion:** Extending the knowledge base and evaluation framework to broader Vietnamese legal domains: criminal procedure, labor law, and land use regulations.
- **Adaptive Temporal Weighting:** Replacing the fixed $\lambda = 0.20$ decay parameter with a learned temporal weighting function adapting to specific legal topics.

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